

Domain and Range from Problem Situations - Practice

Algebra II Honors | Unit 1 | Lesson 2 Practice | TEKS A2.7.I, A2.2.A

Objective: I can build a function from a described situation, find its contextual domain, and use average rate of change to interpret it.

Section 1 — Recall

1. A club charges \$25 annual dues plus \$4 per event attended, up to 20 events. Write the cost function $C(n)$ and give its contextual domain.

 $C(n) =$ _____ Domain: _____

2. A tank drains at a constant rate from 50 gallons to 0 over 10 minutes. Write the volume function $V(t)$ and give the domain and range.

 $V(t) =$ _____ Domain: _____ Range: _____**Section 2 — Apply**

3. A rectangle has perimeter 50 ft, with one side x . Build $A(x)$ and give its contextual domain, justifying why it is open at both ends.

 $A(x) =$ _____ Domain: _____ — _____

4. A ball is launched from ground level: $h(t) = -16t^2 + 80t$ feet, until it lands. Find the contextual domain (show the factoring) and the range.

Factoring: _____ Domain: _____ Range: _____

Section 3 — Challenge

5. A company's profit is $P(n) = 8n - 120$ dollars for n units sold, $0 \leq n \leq 50$. Find the average rate of change of P from $n = 10$ to $n = 30$, and explain in one sentence what the number means for the business.

Rate: _____ Meaning: _____

6. A ball's height is $h(t) = -16t^2 + 32t + 5$ feet. Find the average rate of change from $t = 0$ to $t = 1$, and from $t = 1$ to $t = 2$, then explain the sign change in context.

On $[0,1]$: _____ On $[1,2]$: _____ Meaning: _____

Section 4 — Explain It

7. A classmate says: "If a contextual domain is an open interval like $(0, 20)$, that just means the function itself is undefined at the endpoints." Explain what's wrong, distinguishing the function's own domain from the domain the situation restricts to.
